

STILLSTACK · PROBLEM STATEMENT

StillStack problem statement

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.5	2026-09-26	Amish Chadha	Budget top-up approved by Amish
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	Record the 1 m2 aperture decision (SSK-DDR-001) and the TRL 3 cost finding; partner question stays open
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (users, context, constraints, out of scope, prior work)
0.1	2026-09-24	Amish Chadha	Initial scaffold

StillStack problem statement

Solar stills are the simplest way to turn salty or brackish water into drinking water without fuel or grid power, but a single-basin still makes only about 3 to 5 L per m² per day, so a household needs 3 to 5 m² of glazed basin to cover drinking and cooking water. That area is too large, too costly and too hard to maintain for most of the people who need it.

■ The problem

A single-basin still throws away almost all of the energy it collects. Sunlight evaporates water from a shallow basin, the vapor condenses on the glass cover, and the latent heat released there (about 2.26 MJ per kg of water) is lost to the sky. The result is a daily efficiency of roughly 30 to 45 % of the solar input, which caps output at about 3 to 5 L per m² per day under good sun. The deep water basin also takes hours to warm up each morning.

Multi-effect distillation fixes this in industry by condensing the vapor from one effect against the evaporator of the next, so the same heat evaporates water several times. Research groups have shown that the same idea works passively at small scale: thin wicks pressed against stacked metal plates, heated by sunlight from above, each stage reusing the heat of condensation from the stage above it. Laboratory prototypes of this kind (for example the passive multistage devices reported by Chiavazzo and colleagues in *Nature Sustainability* in 2018 and the thermally localized multistage still reported by Xu and colleagues in *Energy and Environmental Science* in 2020) produced two to four times the output of a basin still per unit area under simulated or real sun. There is still no open, documented, garage-buildable design that a community workshop, NGO or school can build, audit and adapt.

■ Users and context

User	Need	Context
Household in an arid or saline coastal area	5 to 15 L per day of safe drinking and cooking water from seawater or brackish well water	Rooftop or yard, no reliable grid, strong sun (5 to 7 kWh per m ² per day)
Rural clinic or school	A steady supply of low-salt water for drinking, and for rinsing and battery top-up	Small compound, a caretaker with basic tools, part-time maintenance
Small island or remote outpost	Backup water when a reverse osmosis unit or its power fails	Seawater feed, salt spray, high UV
NGO, makerspace or technical college	A reproducible, inspectable design to build, test and improve	Workshop with hand tools, a 3D printer and sheet metal tools

■ Constraints

- Garage-buildable prototype of 1 m² aperture for about \$355 USD in parts, using sheet aluminum, polycarbonate glazing, timber and cloth wicks. The 1 m² aperture was confirmed by Amish on 2026-09-25 (SSK-DDR-001), the budget was raised from \$250 to \$320 the same day (SSK-DDR-002), and Amish approved a top-up to \$355 on 2026-09-26. The priced BOM, including the 150 °C materials and stagnation cover decided in SSK-DDR-002, comes to about \$351, so the cost constraint is met (SSK-CAL-001).
- Fully passive: no pumps, electronics or batteries. Feed water arrives by gravity from a container the user fills.

- Every surface that touches distillate must be food-contact safe, and all materials must tolerate dry stagnation under full sun.
- Feed water may be seawater (about 35 g per L of salt) or brackish groundwater, so wicks and plates must resist salt build-up and corrosion.
- Maintenance by a non-specialist with no special tools: rinsing or replacing wicks, clearing salt.
- Distillate is a raw product. It is not assumed to be safe to drink until tested, and the documentation must say so.

■ Out of scope

- Certified drinking water treatment or any claim of potability.
- Brine disposal or salt recovery beyond a drain outlet.
- Large community plants (more than about 10 m²) and powered or hybrid (photovoltaic plus heater) versions.
- Water quality testing equipment and remineralization systems, which are named as needs but not designed here.

■ Open questions

- Which users to involve first, and through which partner (a coastal NGO, a university water lab or a makerspace network)? No recommendation has been made. Proposed, awaiting Amish.
- The first size is settled: a household unit of 1 m², decided by Amish on 2026-09-25 (SSK-DDR-001).